WHICH TOOL? MATCH WHAT THE QUESTION SAYS

Classify a system (linear, TI, causal, memoryless, stable, real)	Past paper: System classification
Block diagram with delays and gains	Past paper: Block diagram to equation
Anti-aliasing / reconstruction / sketch spectrum	Past paper: Sampling questions
Two-stage output with H(s) or H(z) and a given input	Past paper: Transfer function questions
Draw or simplify $x(-t+2)$, $x(2t)$, $x[n-3]$	W1 Transformations: reflect first, then shift
Is this signal periodic? Find T_o or N_o	W1 Periodicity (DT needs $2\pi/\omega_o$ rational)
Linear? Time-invariant? Causal? Stable? Memory?	W2 Property tests, then W3 table if you have h
Find y given x and h (or h given the equation)	W3 Convolution (flip h, then shift)
Periodic input, find output	W6: $Y_k = H(k\omega_o)X_k$
Input is a sinusoid or $e^{j\omega t}$	W5: output = $\lvert H(\omega)\rvert\cos(\omega t+\varphi+\angle H(\omega))$
Aperiodic input, signal spectrum, Parseval	W8 CTFT / DTFT
Sampling rate, aliasing, reconstruction	W9: $T_s < \pi/W$
Differential equation, RoC, poles, stability (CT)	W10–W11 Laplace, H(s)
Difference equation, RoC, poles, stability (DT)	W12 Z-transform, H(z)
Give the transform of a signal	Always state the RoC with it (Laplace, Z)
Output with initial rest or step input	Y = HX, then partial fractions, RoC outside poles if causal

TOPICS

Exam format and marking (2022, 2024, sample paper)

Use when: Read once before you start.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
2024 exam: unlimited physical notes, a physical formula sheet and a physical calculator are allowed	No second device. Paper copies only.	Print this sheet. The pages are set up for print.
The exam gives you tables: CT and DT Fourier series, CTFT, DTFT, Laplace, Z (pairs and properties)	Look the pairs up. Do not memorise them.	Spend notes space on methods and conditions, which are not in the tables.
Section A: select or type answers (1 mark each). Section B: handwritten, reasoning required	"No marks will be given if no reasoning, formulae or calculations are provided."	
Marks tell you how much to write: 2 marks = answer + one line of reason; 3 marks = answer + reason + one equation	Sample paper marking: 0.5–1 mark for the answer, the rest for the explanation.	A correct number with no explanation scores about 1 of 4.
Every paper has: system properties, a sampling question, a transfer function with RoC and stability, a Fourier series question, and a DT difference equation question		

Exam

What the formula sheet gives you and what it does not

Use when: Checked against Formula_Sheet.pdf (7 pages).

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Given: sifting, convolution (DT and CT), geometric sum $1/(1-\alpha)$, integrals of e^x and x^k, $X_s(\omega) = (1/T)\Sigma X((\omega-2\pi k)/T)$		
Given: CT and DT Fourier series tables, CTFT and DTFT tables, Laplace and Z tables with properties	Pairs and properties only.	No RoC rules, no partial-fraction recipe, no stability tests.
Not given: system property tests, periodicity rules, aliasing condition $T_s<\pi/W$, ideal filter responses, Parseval tricks	These are what this sheet is for.	
Laplace table has NO left-sided pair	Only right-sided signals: δ, u, tu, $e^{-at}u$, $te^{-at}u$, $e^{-bt}\cos/\sin(\omega_o t)u$.	You must supply $-e^{-at}u(-t) \Leftrightarrow 1/(s+a)$, $Re(s)<-Re(a)$ yourself. Also $\cos(\omega_o t)u$ and $\sin(\omega_o t)u$ are the b=0 cases.
Z table includes $\alpha^n u[-n]$ ($\|z\|<\|\alpha\|$) but the left-sided $u[-n-1]$ form is not listed	Left-sided: $-\alpha^n u[-n-1] \Leftrightarrow 1/(1-\alpha z^{-1})$, $\|z\|<\|\alpha\|$. Write it from your notes.	
DT Fourier series table has a typo in the square-wave row (see Fourier series card)	Also the DT Parseval line prints "dt". Read it as a sum.	
CT Fourier series square wave: $\sin(k\omega_0 T)/(k\pi)$ for $u(t+T)-u(t-T)$	Given.	$X_o = 2T/T_o$ is not listed. Take the limit $k\to 0$ of the formula.

Exam System classification

Use when: Seen: 2022 Q2, 2024 Q1, mid-sem Q1, sample and mock.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Memoryless? Only the current input appears, with the same time index	x(t+2), x[n−1], sums over k all have memory.	
Causal? Any input at a later time than the output?	Memoryless ⇒ causal.	Σ over k=−1..1 of x[n−k] uses x[n+1], so non-causal. $e^{-t}x(t+2)$ is non-causal.
Linear? Check $x \to 0$ gives 0, then scaling and addition	A fixed multiplier of the signal is fine (e^{-t}, sin t, n^2).	Squaring x, adding a constant, or |x| all break linearity. Initial rest keeps y=x+0.5x(t−1) linear.
Time-invariant? Does any coefficient depend on t or n?	Time-varying multiplier ⇒ TV.	$e^{-t}x(t+2)$ is TV. $2e^{j\omega_1 t}x(t-1)$ is TV. $x[n-1]^2+x[n]$ is TI but non-linear.
BIBO stable? Bound |y| using $\|x\| \le M$	Finite sum of delayed copies ⇒ stable.	Complex exponential multipliers have magnitude 1, so they stay stable.
Real system? Real input must always give real output	Complex coefficients or $e^{j\omega t}$ multipliers ⇒ not real.	2024 Q1: $y = 2e^{j\omega_1 t}x(t-1)+e^{j\omega_2 t}x(t-2)$ is linear, TV, causal, has memory, stable, NOT real.
"Is it LTI?" for a system that creates new frequencies	An LTI system only scales and phase-shifts each input frequency.	Mock Q6d: 440 Hz tone in, 330 and 550 Hz out ⇒ not LTI.

Stuck? Try: Write y for x=0, for 2x, and for x_1+x_2, then for a delayed x. Four lines, four answers.

Exam Sampled values and DT sequences

Use when: Seen: 2022 Q1, 2024 Q2.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$x_s[n] = x(nT_s)$. Build a table of $t = nT_s$ first	Fill n=0..5 in order.	cos(2πt) at 10/s: 1, 0.81, 0.31, −0.31, −0.81, −1.
sin(10πt) at 10 samples/s	$\omega_o T_s = \pi$.	All samples are 0 (critical sampling).
u(t−0.25) at t=0,0.1,...,0.5	0, 0, 0, 1, 1, 1.	
sin(2πt)u(t) at 10/s	0, 0.59, 0.95, 0.95, 0.59, 0.	
z(t)=u(t+1)−u(t−1), T_s=2/3: x[n]=1 for $\|nT_s\| < 1$, i.e. $n = -1, 0, 1$	Evaluate at nT_s, n = −2, −1, 0, 1 gives 0, 1, 1, 1.	Then $y[n] = \Sigma_{k=-1}^{1} h[n-k]$: A=−1, B=1.

Stuck? Try: Draw the CT signal and mark the sample instants nT_s on the axis.

Exam Block diagram to difference equation

Use when: Seen: 2024 Q2c, mock Q2, sample Q2.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Label each wire with its Z-transform. A delay multiplies by z^{-1}	Do the algebra in the z-domain.	Gains multiply the wire they sit on. Adders add the arriving wires.
Collect: $\hat{Y} = \Sigma(\text{gain} \times \text{delays})\ \hat{X} + \text{feedback} \times \hat{Y}$, then read each coefficient	Read the shared final gain (0.1 in 2024 Q2c) into every term.	
2024 Q2c: $y[n]=0.5y[n-1]+x[n]+0.2x[n-1]+0.1x[n-2]$	Matching coefficients gives a=1, b=2, d=0.5 with c=10 and final gain 0.1.	
Direct difference equation to FIR diagram: $y[n] = \Sigma\ b_k\ x[n-k]$	Delay chain, tap gains b_k, one adder.	$h[n] = \Sigma\ b_k\ \delta[n-k]$ (mock Q1).

Exam Read h or the transform from a given expression

Use when: Seen: sample Q4, mock Q1, mid-sem (impulse response of deltas).

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Impulse response of an FIR equation: replace x by δ	LTI.	$h[n] = \delta[n] - 0.5\delta[n-1] - 0.5\delta[n-2] + \delta[n-3]$.
$H(\omega) = 1.5 + 0.5\cos\omega + \cos 2\omega$: rewrite with Euler, then the coefficient of $e^{-j\omega n}$ is h[n]	$\cos\omega = \frac{1}{2}e^{j\omega} + \frac{1}{2}e^{-j\omega}$.	h[−2]=0.5, h[−1]=0.25, h[0]=1.5, h[1]=0.25, h[2]=0.5. Not causal.
If $H(\omega)$ is multiplied by $e^{-j\omega N}$, the whole h shifts right by N	Delay property.	Multiply by $e^{-j2\omega}$: all indices ≥ 0, so the system becomes causal.
$h(t) = \delta(t-1) - \delta(t-2) + \delta(t-3)$	Finite number of impulses ⇒ $\int\lvert h\rvert = 3 < \infty$.	Stable. Step response $s(t) = u(t-1) - u(t-2) + u(t-3)$. Output = $x(t-1) - x(t-2) + x(t-3)$.
Sketch the output of a delta-sum system for a pulse input	Delay the pulse by 1, 2, 3 and add with signs.	The copies overlap: add them point by point.

Exam

Fourier series questions

Use when: Seen: 2022 Q3, 2024 Q4, mid-sem Q2, sample Q8, mock Q3 and Q4.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$X_0 = (1/T_0)\int$ over one period of x = average value	Periodic extension of a finite pulse.	2022 Q3: $x=1-t^2$ on $\|t\|\le1$, $T_o=2$ ⇒ $X_o = 2/3$. 2024 Q4: duty cycle β ⇒ $X_o = \beta$.
$Y_k = H(k\omega_0)\,X_k$. $Y_0 = H(0)\,X_0$	Stable LTI, periodic input.	2022 Q3c: $h = e^{-2t}u - e^{-4t}u$ ⇒ $H(0)=1/2-1/4=1/4$, $Y_o=1/6$.
Parseval: $(1/T_0)\int\|z\|^2 = \Sigma_{k=-\infty}^{\infty}\|Z_k\|^2$. Real z: $\Sigma_{k=1}^{\infty}\|Z_k\|^2 = (\text{power} - \|Z_0\|^2)/2$	Real, periodic.	Mid-sem: power 4, $Z_o=0$ ⇒ $\Sigma_{k\ge1}\|Z_k\|^2 = 2$.
Which harmonics survive a filter? Find $k\omega_0$ for each k and compare with the passband	Include negative k.	Sample Q8: $T=2\pi$ ⇒ only $k=\pm1$ passes ⇒ $y = 2\|X_1\|\cos t$. $T=4\pi$ ⇒ $k=\pm2$ passes. $T=8\pi$ ⇒ several pairs ⇒ not a single cosine.
Real pair X_k, X_{-k} both in the passband ⇒ $A\cos(k\omega_0 t+\angle X_k)$ with $A = 2\|X_k\|$	Real signal.	
Modulation: $\cos(\omega_c t)x$ ⇒ $W_k = \tfrac{1}{2}X_{k-1} + \tfrac{1}{2}X_{k+1}$ when $\omega_c = \omega_0$	ω_c is an integer multiple of ω_o.	Demodulator (mock Q4): choose A so $Y_o = A\cdot W_o$ matches, and $0 < B < \omega_c$ to block $k\ne0$. Linear but not time-invariant.
Fundamental frequency of a sum of sinusoids = gcd of the component frequencies	Mock Q6a: 440, 330, 550 Hz ⇒ 110 Hz, so $\omega_o = 220\pi$.	
Differential equation from H(ω): cross-multiply, then $j\omega \to d/dt$	Initial rest.	2024 Q4a: $(1+ja\omega-\omega^2)Y = (1-(\omega/2\pi)^2)X$ ⇒ $y + a\,y' + y'' = x + x''/4\pi^2$.
If $H(k\omega_0)=0$ at some harmonic, x is not unique from y	Add or remove that harmonic.	2024 Q4d: add $10\cos(2\pi t+\varphi)$ to x when $H(2\pi)=0$.

Stuck? Try: Write ω_o, list which k are relevant, evaluate $H(k\omega_o)$ for each. Then build y.

Exam Frequency response and sinusoid questions

Use when: Seen: 2022 Q4, sample Q6, mock Q6.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Output sinusoid → input: divide the output phasor by $H(\omega_0)$	Real stable system.	Do each frequency separately.
Input constant a ⇒ output a·H(0)	$x = a\,e^{j0t}$.	Sample Q6: H(ω)=Y/X, then y = a·H(0) = 1.
Periodic input: $Y_k = H(k\omega_0)X_k$	Use $\omega_o = 2\pi/T_o$.	
Delay + gain system: $h = 0.5\delta(t-0.1)$, $H(\omega) = 0.5e^{-0.1j\omega}$	Undo it: z(t) = 2y(t+0.1).	
$H = Y/X$ using FT tables	If X(ω) = 0 anywhere, H is undetermined there.	

Exam Fourier transform algebra

Use when: Seen: 2022 Q5, 2024 Q7, mock Q3.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Improper rational (4+jω)/(5+jω): divide first $= 1 - 1/(5+j\omega)$	Degree numerator = degree denominator.	The constant 1 gives δ(t). y(t) = δ(t) − $e^{-5t}u(t)$.
$X = Y/H$, then simplify and look up each term	H(ω) ≠ 0.	2024 Q7: X = (2+jω)/(5+jω) = 1 − 3/(5+jω) ⇒ x = δ(t) − $3e^{-5t}u(t)$.
$n\alpha^n u[n] = (n+1)\alpha^n u[n] - \alpha^n u[n]$ ⇒ $1/(1-\alpha e^{-j\omega})^2 - 1/(1-\alpha e^{-j\omega})$	Table has $(n+1)\alpha^n u[n]$ and $\alpha^n u[n]$.	2022 Q5a with α=0.2.
$0.2^n \cos(n)\,u[n]$: write $\cos n = \tfrac{1}{2}e^{jn} + \tfrac{1}{2}e^{-jn}$, then modulation	Frequency shift by ±1.	$\tfrac{1}{2}/(1-0.2e^{-j(\omega-1)})$ + $\tfrac{1}{2}/(1-0.2e^{-j(\omega+1)})$.
$\cos(\omega_0 t)x(t)$ ⇒ $\tfrac{1}{2}X(\omega-\omega_0) + \tfrac{1}{2}X(\omega+\omega_0)$	Mock Q3: cos(aπt)x with 10 periods in length 2 ⇒ a = 10, X(ω)=2 sin ω/ω.	
Use the table you are given	Current formula sheet: u[n+N] −u[n−N] ⇔ $[\sin(N\omega)/\sin(\omega/2)]e^{j\omega/2}$ (2N samples, n=−N..N−1).	The 2024 paper printed $\sin(\omega(N+\tfrac{1}{2}))/\sin(\omega/2)$, which is the (2N+1)-sample window u[n+N] −u[n−N−1]. Count the samples before choosing.

Exam

Sampling, aliasing, reconstruction questions

Use when: Seen: 2022 Q6, 2024 Q6 and Q10 (sample), mock Q8, sample Q1.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
No aliasing iff sampling frequency $> 2 \times$ highest frequency ($T_s \cdot W < \pi$)	Sinusoid: sin(4πt) has f=2 Hz, so f_s>4.	Bandpass signal: use the highest edge W, not the band width. 5π–8π ⇒ W=8π ⇒ Nyquist rate 8 samples/s.
Anti-aliasing LPF cutoff $\le f_s/2$ Hz $= \pi f_s$ rad/s	Ideal LPF h = sin(Wt)/(πt) has H = u(ω+W) − u(ω−W), cutoff W.	Sample Q1: largest W = π f_s. f_s=500 ⇒ 500π rad/s.
Hz to rad/sample: $\omega = 2\pi f/f_s$	DT filter design.	Sample Q10: 50 Hz at f_s=500 ⇒ π/5. High-pass edge at π/5. Sketch −2π to 2π with edges ±π/5, ± (2π−π/5).
$X_s(\omega) = (1/T_s)\,\Sigma\, X((\omega - 2\pi\ell)/T_s)$. Inverse: $X(\omega) = T_s\, X_s(T_s\omega)$ inside the band	Mock Q8: peak 20 → X peak 2, edge (2π/3)/0.1 = 20π/3.	Horizontal axis scales by $1/T_s$, vertical by T_s.
Energy of x_s: $\Sigma\lvert x_s[n]\rvert^2 = (1/2\pi)\int$ over one period of $\lvert X_s\rvert^2$	Parseval.	Mock Q8b = 800/9. Use evenness to integrate over 0 to π and double.
Impulse-train sampling $p(t)=\Sigma\delta(t-nT)$: $Z(\omega) = (1/T)\,\Sigma\, X(\omega - 2\pi k/T)$	T = 1/4 ⇒ ω_s = 8π, copies every 8π.	2024 Q6: H_2 = T = 1/4 on the band, 0 elsewhere. H_1 = 1 or a band-limiting filter.
ZOH: output is held; reconstruction delay ≈ $T_s/2$	Order of ideal filter and sampler matters only if the signal is not already band-limited.	
Always say why: "sampled at f_s above twice the highest frequency, so the DTFT copies do not overlap"	Marks are for the sentence.	

Exam Transfer function, RoC, stability, output

Use when: Seen: 2022 Q7, 2024 Q8, mock Q5 and Q7, sample Q5.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Block diagram: name wires, write equations in s, solve for $\hat{Y}/\hat{X}$	Integrator block 1/s, delay $e^{-s\tau}$, gain constants.	2024 Q8: $\hat{H} = (2s^2-1)/((s-1)(s+2))$.
RoC: "all subsystems causal" ⇒ right of the rightmost pole	Rational $\hat{H}$.	2024 Q8: poles 1 and −2 ⇒ RoC Re(s) > 1.
Stable? RoC must contain the jω axis; equivalently all poles in the open left half-plane	Pole at +1 ⇒ not stable. Poles at ±2j (sample Q5) ⇒ not stable.	
Output for a given input: $\hat{Y} = \hat{H}\hat{X}$, partial fractions, inverse with the right-sided table	Causal system, right-sided signals.	$te^{-t}u+e^{-t}u$ has $\hat{X} = (s+2)/(s+1)^2$. It cancels the (s+2) pole.
Repeated pole at −1: $\hat{Y} = A/(s+1)^2 + B/(s+1) + C/(s-1)$	Get C by cover-up, A by cover-up, B by evaluating at s=0 (or any s).	2024 Q8e: $y = -\frac{1}{2}te^{-t}u + (7/4)e^{-t}u + (1/4)e^{t}u$.
Step response: inverse of $\hat{H}(s)/s$	Causal.	Sample Q5: $\hat{H} = 2s/(s^2+4)+s/(s+1)$ ⇒ $y = [\sin 2t + e^{-t}]u(t)$.
Bounded step response does NOT imply BIBO stability	Same example: bounded step response, poles on the jω axis.	
ODE $y''+ay'+by=x$ ⇒ $\hat{H} = 1/(s^2+as+b)$. Causal and stable ⇒ poles with negative real part	Pick p=−1, q=−2 ⇒ s^2+3s+2 ⇒ a=3, b=2.	
$\hat{Y}=1/(s(s-1))$, RoC Re(s)>1 ⇒ $y = (e^t-1)u(t)$	Right-sided inverse for both terms.	$\hat{H} = \hat{Y}/\hat{X} = (s+2)/(s(s-1))$ when $\hat{X}=1/(s+2)$ (RoC Re(s)>−2).

Exam Poles, zeros and properties puzzle

Use when: Seen: sample Q3. Every property in the statement is a clue.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
"Input 1 for all t gives output 0" ⇒ $H(0) = 0$ ⇒ zero at $s = 0$	Constant = e^{0t}.	
"Real-valued system" ⇒ complex poles and zeros come in conjugate pairs	Pole at −1+j ⇒ also −1−j.	
"RoC is Re(s) > 2" ⇒ rightmost pole has real part 2	Three poles, two used by the pair, so the third is real: s = 2.	
Right-sided RoC ⇒ right-sided impulse response. Strip −1<Re(s)<2 ⇒ two-sided		
Write each answer with the property that forced it	The hint says: refer to all properties.	1 mark for the location, 1 for the reason.

Exam Z-domain questions

Use when: Seen: 2022 Q8, 2024 Q3 and Q5, mock Q9, sample Q7.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$y[n]+2y[n-1]+y[n-2] = \frac{1}{2}x[n]+x[n-1]-x[n-2]$: $\hat{H} = (\frac{1}{2}+z^{-1}-z^{-2})/(1+2z^{-1}+z^{-2})$. Multiply by $2z^2$: $(z^2+2z-2)/(2z^2+4z+2)$	To match the form $(z^2+az+b)/(cz^2+dz+e)$: a=2, b=−2, c=2, d=4, e=2.	Initial rest ⇒ causal and LTI.
Poles: solve $z^2+z+1=0 \Rightarrow z = -\frac{1}{2} \pm j\sqrt{3}/2 = e^{\pm j2\pi/3}$	Polar form means magnitude and angle.	$\lvert z\rvert=1$: on the unit circle.
Causal ⇒ RoC $\lvert z\rvert > 1$. Unit circle not inside ⇒ not BIBO stable	Poles on the circle are not stable.	
$h[n]$ for $(1+0.5z^{-1})/(1+z^{-1}+z^{-2})$: match $\alpha^n\cos(\omega_0 n)u[n]$ with $\alpha=1$, $\omega_0=2\pi/3$ ($-\cos\omega_0 = 0.5$)	$h[n] = \cos(2\pi n/3)u[n]$.	
Unbounded output for a bounded input: a real bounded x at the pole frequency	Resonance.	$x[n]=\cos(2\pi n/3) \Rightarrow$ y grows. Say "bounded by 1, output unbounded".
Given $x[n]=(0.5)^n u[n]$, $y[n]=(0.25)^n u[n]$: $\hat{H} = \hat{Y}/\hat{X} = (1-0.5z^{-1})/(1-0.25z^{-1})$	Table pairs for $\alpha^n u[n]$.	IIR (pole at 0.25 ≠ 0). $h = 2\delta[n] - 0.25^n u[n]$.
FIR ⇔ no poles except z=0. IIR ⇔ a pole elsewhere		
Step response in Z: $\hat{H}(z)/(1-z^{-1})$, RoC $\lvert z\rvert>1$	Causal.	Sample Q7: $1/((1+z^{-1}+z^{-2})(1-z^{-1})) = 1/(1-z^{-3})$.
Write the difference equation from $\hat{H}$: cross-multiply, z^{-m} → delay by m	State "subject to initial rest".	Missing initial rest loses a mark.
2022 Q8: $y[n]=x[n]+2x[n+1]-2y[n-1]$: $x[n+1]$ gives a positive power of z in the numerator	Pole at −2, so RoC $\lvert z\rvert>2$ for a causal solution.	Not stable (pole outside the unit circle).

Stuck? Try: Always: transfer function → poles → RoC from causality → stability test → output.

W1 Transformations of signals

Use when: x(t−τ), x(αt), x(−t+2), "draw y" questions.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$y(t) = x(t-\tau)\ y[n] = x[n-N]$	τ>0 or N>0 shifts right (delay); negative shifts left (advance).	"+" inside the bracket moves the graph LEFT. Check one point.
$y(t) = x(\alpha t)$	|α|>1 compresses, |α|<1 stretches. α=−1 is time reversal.	Dilation is clean for CT only; DT x[n/2] is undefined at odd n.
$x(-t+2) = x(-(t-2))$	Reflect first: z(t)=x(−t), then shift right by 2: y(t)=z(t−2).	Shift-then-reflect gives the wrong picture. Check y(0)=x(2), y(4)=x(−2).
$y = \alpha x,\ y = x_1+x_2,\ y = x_1x_2$ (pointwise)	α real or complex.	

Stuck? Try: Plug in two or three t values and read them off x. Do not trust the order of operations by eye.

W1 Complex numbers, exponentials, sinusoids

Use when: Magnitude, phase, growth or decay, writing x[n] from a plot.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$x(t) = x_R + j\,x_I,\ r = \sqrt{(x_R^2 + x_I^2)},\ x = r\,e^{j\theta}$	Phase usually in (−π, π].	$\tan^{-1}$ alone loses the quadrant. Use the case rule for x_R<0.
$e^{j\omega_0 t} = \cos\omega_0 t + j\sin\omega_0 t,\ \lvert e^{j\omega_0 t}\rvert = 1$	ω_o>0 rotates anticlockwise.	
General CT: $e^{st} = e^{\sigma t}e^{j\omega t},\ s = \sigma + j\omega$	σ<0 decays, σ>0 grows.	
General DT: $\alpha^n = r^n e^{j\omega_0 n},\ \alpha = re^{j\omega_0}$	r<1 decays, r>1 grows.	Read x[0] first to get the amplitude, then r from x[1]/x[0].
$A\cos(\omega_0 t+\theta) = A\ \mathrm{Re}\{e^{j(\omega_0 t+\theta)}\}$	A>0.	DT: $\lvert\omega_o\rvert$>π looks like a smaller frequency (aliasing).

Stuck? Try: For a decaying DT plot: x[0]=β, x[1]=βr gives r. A real positive plot means ω_o=0.

W1 Unit step and unit impulse

Use when: Build a pulse train from steps, sifting integrals and sums.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$u[n]=1$ for $n\geq 0$, else 0. $u(t)=1$ for $t>0$, 0 for $t<0$	In this unit $u(0)$ is left undefined in CT.	
$\delta[n]=1$ only at $n=0$. $\Sigma\, x[n]\delta[n] = x[0]$		
$\int x(t)\delta(t)\, dt = x(0)$	x continuous at 0.	
Pulse from t_1 to $t_2 = u(t-t_1) - u(t-t_2)$	Repeated pulses: find one, then shift copies.	
$\delta(t) \approx [u(t+\varepsilon) - u(t)]/\varepsilon$ as $\varepsilon \to 0$	Area stays 1.	

Stuck? Try: Use $\delta(t-t_o)$ inside an integral to pick $x(t_o)$. Delta and step relate by derivative or integral.

W1 Periodicity

Use when: "Is it periodic? Find the fundamental period."

FORMULA / FACT	CONDITIONS NEEDED	TRAP
CT: $x(t+T)=x(t)$. $e^{j\omega_0 t}$ has $T_0 = 2\pi/\lvert\omega_0\rvert$	Always periodic for $\omega_o \neq 0$.	
DT: $x[n+N]=x[n]$, N a positive integer. $N = m\cdot(2\pi/\lvert\omega_0\rvert)$	Periodic iff $2\pi/\omega_o$ is rational. Take the smallest m that makes N an integer.	$\sin(4\pi n/5)$: $2\pi/\omega_o=5/2$, so m=2, $N_o=5$. Fundamental frequency is $2\pi/N_o$, not the given ω.
Fundamental frequency: $f_0=1/T_0$, $\omega_0=2\pi/T_0$ (CT), $\omega_0=2\pi/N_0$ (DT)		
Sum of two periodic signals	Same period gives that period. DT sum is always periodic. CT sum with different periods only if the ratio is rational.	Sampling a periodic CT signal may give an aperiodic DT signal.

Stuck? Try: Write $\omega_o/2\pi$ as a reduced fraction a/b. Then N_o = b. Irrational means aperiodic.

W1 Bounded, even, odd

Use when: Bounded signal tests, symmetry shortcuts.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Bounded: $\exists M, \lvert x(t)\rvert \le M$ for all t	Use the magnitude for complex signals.	$\lvert e^{j(\pi/2)n + \cos n}\rvert = e^{\cos n}$, so bounded with M = e.
Even: $x(-t)=x(t)$. Odd: $x(-t)=-x(t)$		
$x_{even} = [x(t)+x(-t)]/2$, $x_{odd} = [x(t) - x(-t)]/2$	Any signal splits this way.	
Odd ⇒ $\int$ from $-a$ to a of $x = 0$		
even·even=even, odd·odd=even, even·odd=odd, even+even=even, odd+odd=odd		

W2 Interconnecting systems

Use when: Block diagrams, delay/gain/adder networks.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Series: $S_2(S_1\ x)$	LTI blocks commute.	Order matters for general (non-LTI) systems.
Parallel: $S_1x + S_2x$		
Feedback: write the signal on every wire, then eliminate	DT: put a delay in every loop.	
Series or parallel of linear (or TI, or LTI) systems is linear (or TI, or LTI)	Gain, delay, integrator and differentiator are each LTI.	A network built only from gains, delays and adders is LTI without a proof.

W2 The five property tests

Use when: Classify a system. Do each test separately.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Memoryless: $y(t)=F(t,x(t))$	Output at t uses only x at t.	Delay, advance, integrator and differentiator all have memory.
Causal: no future input	Memoryless systems are causal. Check y at one time.	$y(t)=x(2t)$ is non-causal ($y(1)=x(2)$). A predictor built from past data is causal.
BIBO stable: $\|x\| \le M \Rightarrow \|y\| \le M'$	To disprove, one bounded input with unbounded output is enough.	Integrator is unstable (step in, ramp out). Differentiator is unstable ($x=\sin t^2$).
Linear: $S(x_1+x_2)=Sx_1+Sx_2$ and $S(\alpha x)=\alpha Sx$	Both parts, for complex α. Zero in must give zero out.	$y=x+1$ fails ($\alpha=0$). x^2 and $\|x\|$ fail. $t\cdot x(t)$ and $x(t^2)$ are linear (they act on time, not on the value).
Time-invariant: $S(\text{Delay } x) = \text{Delay}(Sx)$	Compare delay-then-S with S-then-delay.	$\sin(t)\cdot x(t)$ is TV. $x(2t)$ is TV. Linear and TI are independent properties.

Stuck? Try: Linear: try $\alpha=2$ or $x=0$. TI: put in a shifted input and compare. Stable: try a step.

W2 Difference and differential equations, initial rest

Use when: A recursive equation is given with no extra condition.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
FIR: $y[n] = \Sigma\, b_k\, x[n-k]$	Always causal and LTI.	
IIR: $\Sigma\, a_k\, y[n-k] = \Sigma\, b_k\, x[n-k]$	Needs an auxiliary condition.	$y[0]=0$ at a fixed time breaks time-invariance. $y[-1]=1$ breaks linearity.
Initial rest: $x[n]=0$ for $n<n_0 \Rightarrow y[n]=0$ for $n<n_0$	Applies to difference and differential equations (CT: y and its first N−1 derivatives are 0 at t_o).	Initial rest gives a causal LTI system.
Running average: $y[n] = (1/L)\Sigma\, x[n-k]$	Causal, FIR, LTI.	

W3 DT convolution and impulse response

Use when: Find y from x and h, find h from a difference equation.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$x[n] = \Sigma\, x[k]\delta[n-k]$ (sifting)		
$h[n]$ = response to $\delta[n]$	LTI system.	h is one signal. The convolution is the general rule.
$y[n] = \Sigma\, x[k]\, h[n-k] = (x*h)[n]$	LTI. Flip h, then shift.	Never write $x[n]*h[n]$ as a pointwise product of values.
Support of $x*h = [a_1+a_2,\ b_1+b_2]$	Both finite-duration.	Add the endpoints. Length = L_1+L_2-1.
$x*h=h*x$, $(x*h_1)*h_2 = x*(h_1*h_2)$, $x*(h_1+h_2)=x*h_1+x*h_2$, $x*\delta=x$, $x*\delta[n-N]=x[n-N]$		
Series: h_1*h_2. Parallel: h_1+h_2		

Stuck? Try: Three ways: table or sliding window, polynomial multiplication (FIR only), or the sum with case splits (IIR).

W3 CT convolution

Use when: Integral convolution with piecewise signals.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$y(t) = \int x(\tau)\, h(t-\tau)\, d\tau$	LTI. Flip h, slide by t, integrate the overlap.	Split into cases for each overlap region. The result should be continuous at the boundaries (no impulses).
$x*\delta = x$, $x*\delta(t-\tau_0) = x(t-\tau_0)$		
Step response $s(t) = (h*u)(t)$; $h(t) = ds/dt$	Step response differentiable.	
Support $[a_1+a_2,\ b_1+b_2]$		

W3 LTI properties from the impulse response

Use when: "Is this LTI system causal / stable / memoryless?" and you are given h.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Memoryless	h[n]=0 for n≠0, h(t)=0 for t≠0	Memoryless LTI = gain system, h = αδ.
Causal	h[n]=0 for n<0, h(t)=0 for t<0	
BIBO stable (DT)	Σ|h[n]| < ∞ (absolutely summable)	1/n decay is NOT summable. u[n] is not summable.
BIBO stable (CT)	∫|h(t)| dt < ∞ (absolutely integrable)	$e^{-t}u(t)$ is integrable (=1). u(t) and (1/t)u(t−1) are not.
Bounded and zero outside [a,b] ⇒ absolutely integrable	Stable does not imply causal (a boxcar around 0 is stable, non-causal).	

Stuck? Try: Stability is the hardest test: check the sum or integral actually converges.

W4 Periodic extension and finite-duration signals

Use when: A finite-duration x is repeated every T.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$x_{per}(t) = \Sigma_m\, x(t - mT)$	Take T = duration b−a (CT) or N_o = B−A+1 (DT).	DT length has the "+1". CT length does not.
Fundamental period of the extension can be shorter than T	Happens when x repeats inside itself.	Two equal triangles in a 2-second window give T_o=1, not 2.
DT piece of length 1 (single impulse)	Its periodic extension is a constant.	

W4 Fourier series formulas

Use when: Periodic x, "find the coefficients", spectrum plots.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
CT synthesis: $x(t) = \Sigma_k X_k e^{jk\omega_0 t}$	Periodic, $T_o = 2\pi/\omega_o$.	
CT analysis: $X_k = (1/T_0) \int$ over one period of $x(t) e^{-jk\omega_0 t} dt$	Any period works.	Choose the interval that uses symmetry.
DT synthesis: $x[n] = \Sigma$ for $k=0..N_0-1$ of $X_k e^{jk\omega_0 n}$	$\omega_o=2\pi/N_o$.	
DT analysis: $X_k = (1/N_0) \Sigma$ for $n=0..N_0-1$ of $x[n] e^{-jk\omega_0 n}$	DT X_k is periodic in k with period N_o. CT X_k is not.	
Read X_k straight from Euler: $\cos(\omega_0 t)=\frac{1}{2}e^{j\omega_0 t}+\frac{1}{2}e^{-j\omega_0 t}$	Coefficients are unique.	No integral needed if x is already a sum of harmonics.
X_0 = average value of x		
DT square wave x=1 for $\lvert n\rvert \le N$ (2N+1 samples): $X_k = \sin(k\omega_0(N+\frac{1}{2})) / (N_0 \sin(k\omega_0/2))$	Check by summing the geometric series.	The formula sheet prints $\sin(k\omega_o(2N+1)/2)/(k\omega_o/2)$, which drops the $1/N_o$ and has no sin in the denominator. Use the corrected form.
Square pulse $\lvert t\rvert<T_1$ in period T_0: $X_k = \sin(k\omega_0 T_1)/(k\pi)$, $X_0 = 2T_1/T_0$	Standard pair. Re-derive if the pulse is shifted.	Shifting by t_o multiplies X_k by $e^{-jk\omega_o t_o}$.
Convergence at a jump → average of the left and right limits	Dirichlet conditions.	Truncated sums overshoot near a jump.

Stuck? Try: Check geometric series first: if $x = \Sigma a^k$-type terms, the coefficients are just those terms. In MATLAB: X = fft(x)/N, x = N*ifft(X).

W4 Fourier series properties

Use when: Modify a signal and ask for new coefficients.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Linearity: $\alpha x+\beta z \Rightarrow \alpha X_k+\beta Z_k$	Same period.	
Real $x \Rightarrow X_k = X^*_{-k}$	$\lvert X_k\rvert$ even in k, $\angle X_k$ odd in k.	
Conjugation: $x^* \Rightarrow X^*_{-k}$		
Modulation: $e^{jM\omega_0 t}x \Rightarrow Y_k = X_{k-M}$	M integer.	$\cos(M\omega_0 t)x$ gives $\frac{1}{2}X_{k-M} + \frac{1}{2}X_{k+M}$.
Periodic convolution $\Rightarrow Y_k = T_0\, X_k Z_k$ (CT); $N_0 X_k Z_k$ (DT)	Convolution over one period.	Do not forget the T_0 or N_0 factor.
Multiplication $\Rightarrow Y_k = \Sigma_{\ell}\, X_\ell Z_{k-\ell}$		
Parseval: average power $= (1/T_0)\int\lvert x\rvert^2 = \Sigma\lvert X_k\rvert^2$	DT: $(1/N_0)\Sigma\lvert x[n]\rvert^2 = \Sigma$ over one period of $\lvert X_k\rvert^2$.	Square the magnitudes. Do not just sum X_k.
Differentiation: $dx/dt \Rightarrow jk\omega_0 X_k$	CT only.	There is no DT differentiation property.

W5 Frequency response

Use when: Response to $e^{j\omega t}$, cos, steady-state, reading H from an equation.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$x = e^{j\omega t} \Rightarrow y = H(\omega)\, e^{j\omega t}$. DT: $x = e^{j\omega n} \Rightarrow y = H(\omega)\, e^{j\omega n}$	BIBO stable LTI. Eternal input (all t).	$y = H(\omega)x$ holds only when x is exactly $e^{j\omega t}$ at that ω.
$H(\omega) = \int h(\tau)e^{-j\omega\tau}d\tau$ (CT); $H(\omega) = \Sigma\, h[k]e^{-j\omega k}$ (DT)	h absolutely integrable or summable.	
$H(\omega) = \lvert H\rvert e^{j\angle H}$. Real system: $H(-\omega) = H(\omega)^*$	$\lvert H\rvert$ even, $\angle H$ odd.	
$x = \cos(\omega t+\varphi) \Rightarrow y = \lvert H(\omega)\rvert \cos(\omega t + \varphi + \angle H(\omega))$	Real, stable system.	Frequency never changes. Amplitude scales by $\lvert H\rvert$, phase shifts by $\angle H$.
DT: $H(\omega) = H(\omega+2\pi)$	Always true.	CT frequency response is generally not periodic.
Delay τ: $H = e^{-j\omega\tau}$. DT unit delay: $H = e^{-j\omega}$	Magnitude 1, linear phase.	
$H(0) =$ DC gain	A high-pass kills a DC offset.	

Stuck? Try: Read H straight off the equation: substitute $x=e^{j\omega t}$, $y=H\, e^{j\omega t}$, divide by $e^{j\omega t}$.

W5 H(ω) from a differential or difference equation

Use when: LCCDE given; find H(ω), |H|, ∠H.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\Sigma a_k (j\omega)^k H(\omega) = \Sigma b_k (j\omega)^k \Rightarrow H = \Sigma b_k (j\omega)^k / \Sigma a_k (j\omega)^k$	CT, stable.	RC circuit: H = 1/(1 + jωRC).
$\Sigma a_k e^{-j\omega k} H(\omega) = \Sigma b_k e^{-j\omega k} \Rightarrow H = \Sigma b_k e^{-j\omega k} / \Sigma a_k e^{-j\omega k}$	DT, stable.	First difference: $H = 1 - e^{-j\omega}$, $\lvert H\rvert = \sqrt{(2-2\cos\omega)}$.
Positive real factor $\times e^{-j\omega \cdot c} \Rightarrow$ magnitude = |real factor|, phase $= -c\omega$	FIR averagers.	Phase jumps by π wherever the real factor changes sign. That is not an error.
Series: H_1H_2. Parallel: H_1+H_2. Feedback: $H_1/(1-H_1H_2)$	Memorise the minus sign.	

W5 Transient and steady state

Use when: Input is switched on: $e^{j\omega t}u(t)$.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$y = H(\omega)e^{j\omega t}u(t) + y_tr(t), y_tr \to 0$	Causal, BIBO stable.	Eternal input has no transient.
FIR ⇒ transient lasts finitely many samples	IIR transients decay but never reach exactly zero.	

W6 Periodic input through an LTI system

Use when: x periodic, find y (CT or DT).

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$Y_k = H(k\omega_0) X_k$	Stable LTI, x periodic with ω_0.	y has the same period as x.
$y(t) = \Sigma H(k\omega_0)X_k e^{jk\omega_0 t}$		
Frequency in x but not in y ⇒ H = 0 there	A frequency missing from both x and y tells you nothing about H.	
Adding a component that sits on a zero of H changes nothing	Check H at every new frequency.	

W6 Ideal filters, FIR design, delay

Use when: Filter type from H, ideal filter sketches, linear-phase FIR.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Ideal low-pass: $\|H\|=1$ for $\|\omega\|<\omega_c$, else 0. $h(t) = \sin(\omega_c t)/(\pi t)$	Symmetric about 0.	Always non-causal. Cannot be built.
DT ideal filters repeat every 2π	Sketch $(-\pi, \pi]$ first, then replicate.	
Series of ideal filters: only the overlap survives. Parallel: passbands add	Non-overlapping passbands in series give the zero system.	
Linear-phase FIR, length L: $\angle H = -((L-1)/2)\omega$	Symmetric coefficients.	Delay is $(L-1)/2$ samples for every passband frequency. Longer filter = better magnitude, more delay.
firpm(L−1, F, A)	First argument is the order = length − 1. F is in units of π.	Needs a transition band. No brick wall.
Reading filter type from $y[n] = x[n] \pm x[n-1]$: sum = low-pass, difference = high-pass	Check H at ω=0 and ω=π.	

W8 Fourier transform definitions

Use when: Aperiodic signal, spectrum, "find X(ω)".

FORMULA / FACT	CONDITIONS NEEDED	TRAP
DTFT: $X(\omega) = \Sigma\, x[k]\, e^{-j\omega k}$. Inverse: $x[n] = (1/2\pi)\int$ over any 2π of $X(\omega)e^{j\omega n}d\omega$	x absolutely summable.	X(ω) is always 2π-periodic.
CTFT: $X(\omega) = \int x(t)e^{-j\omega t}dt$. Inverse: $x(t) = (1/2\pi)\int X(\omega)e^{j\omega t}d\omega$	x absolutely integrable.	Course convention is ω in rad/s with the 1/2π in the inverse.
Geometric series: $\Sigma\, \alpha^n = 1/(1-\alpha)$ for $\|\alpha\|<1$	Spot this before integrating.	
Discrete in one domain ⇔ periodic in the other	Periodic signal ⇒ discrete spectrum. DT signal ⇒ periodic spectrum.	

Stuck? Try: Check whether x is already in the pairs table, then apply properties instead of integrating.

W8 Fourier transform properties (CT and DT)

Use when: Chain properties on a known pair.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Linearity: $\alpha x+\beta z \Rightarrow \alpha X+\beta Z$		
Delay: $x(t-\tau) \Rightarrow e^{-j\omega\tau}X(\omega)$. DT: $x[n-N] \Rightarrow e^{-j\omega N}X(\omega)$		
Modulation: $e^{j\omega_0 t}x \Rightarrow X(\omega-\omega_0)$	Spectrum slides right by ω_0. DT wraps every 2π.	
Convolution: $x*z \Rightarrow XZ$	Standard exam pattern: linearity + delay + convolution in one line.	
Multiplication: $xz \Rightarrow (1/2\pi)\int X(\Omega)Z(\omega-\Omega)d\Omega$	DT integral over any 2π interval.	
Time scaling (CT): $x(at) \Rightarrow (1/\lvert a\rvert)X(\omega/a)$	$a\neq 0$ real.	No general DT version. Only $x[n/k]$ (zeros between) gives $X(k\omega)$.
Conjugation: $x^* \Rightarrow X(-\omega)^*$		
Real $x \Rightarrow X(-\omega) = X(\omega)^*$	$\lvert X\rvert$ even, $\angle X$ odd.	
Even real $x \Rightarrow X$ real		
Parseval: $\int\lvert x\rvert^2dt = (1/2\pi)\int\lvert X\rvert^2d\omega$. DT: $\Sigma\lvert x[n]\rvert^2 = (1/2\pi)\int$ over 2π of $\lvert X\rvert^2d\omega$	Use Parseval to turn a hard integral like $\int 1/(1+\omega^2)d\omega$ into $2\pi\int e^{-2t}dt = \pi$.	
AM: $A\cos(\omega_c t+\theta)x(t) \Rightarrow (A/2)[X(\omega-\omega_c)e^{j\theta} + X(\omega+\omega_c)e^{-j\theta}]$		

W8 Transform pairs (given table)

Use when: Look these up. Combine with properties.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$e^{-at}u(t) \Leftrightarrow 1/(a+j\omega)$	Re(a)>0.	
$te^{-at}u(t) \Leftrightarrow 1/(a+j\omega)^2$	Re(a)>0.	
$u(t+T_1) - u(t-T_1) \Leftrightarrow 2\sin(\omega T_1)/\omega$		
$\sin(Wt)/(\pi t) \Leftrightarrow u(\omega+W) - u(\omega-W)$	W>0.	Ideal low-pass impulse response.
$\delta(t) \Leftrightarrow 1.\ \delta(t-\tau) \Leftrightarrow e^{-j\omega\tau}$		
$u(t) \Leftrightarrow 1/(j\omega) + \pi\delta(\omega)$		
$e^{j\omega_0 t} \Leftrightarrow 2\pi\delta(\omega-\omega_0)$		
DT: $\alpha^n u[n] \Leftrightarrow 1/(1-\alpha e^{-j\omega})$	$\lvert\alpha\rvert<1$.	
DT: $(n+1)\alpha^n u[n] \Leftrightarrow 1/(1-\alpha e^{-j\omega})^2$	$\lvert\alpha\rvert<1$.	
DT window $u[n+N]-u[n-N] \Leftrightarrow [\sin(N\omega)/\sin(\omega/2)]e^{j\omega/2}$	Matches the current formula sheet and your notes (2N samples).	The (2N+1)-sample window $u[n+N]-u[n-N-1]$ gives $\sin(\omega(N+½))/\sin(\omega/2)$, as printed in the 2024 paper.
DT: $\sin(Wn)/(\pi n) \Leftrightarrow 2\pi$-periodic extension of $u(\omega+W) - u(\omega-W)$	$0<W<\pi$.	
DT: $\delta[n-n_0] \Leftrightarrow e^{-j\omega n_0}$		

W8 LTI systems and generalised transforms

Use when: Output of a system, periodic signals in the Fourier transform.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$Y(\omega) = H(\omega)X(\omega)$ (H = FT of h)	Stable LTI.	H = Y/X fails where X=0 (that is 0/0, not H=0).
Periodic x: $X(\omega) = 2\pi\ \Sigma_k\ X_k\ \delta(\omega - k\omega_0)$	X_k are the Fourier series coefficients.	Periodic signals have no ordinary FT (not absolutely integrable).
DT periodic: impulse train repeating every 2π	A single impulse is not enough in DT.	
DC motor and similar: derive $H(\omega)$ from the ODE, read h from the table	Impulse cannot always be measured.	

W9 Sampling and aliasing

Use when: Choose a sampling rate, find an aliased frequency.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$x_s[n] = x(nT_s)$, $\omega_s = 2\pi/T_s$, $f_s = 1/T_s$		
$\cos(\omega_0 t)$ sampled: no aliasing iff $T_s < \pi/\omega_0$	Exactly $T_s = \pi/\omega_0$ is a degenerate case (samples can be all zero). Never use it.	
Bandwidth W: $X(\omega)=0$ for $\lvert\omega\rvert>W$. No aliasing iff $W \cdot T_s < \pi$	Nyquist frequency = π/T_s.	Nyquist rate is 2W (a rate). Nyquist frequency is π/T_s. Do not mix them up.
$X_s(\omega) = (1/T_s)\,\Sigma_\ell\, X((\omega - 2\pi\ell)/T_s)$	DTFT of the samples.	Copies of X scaled by T_s horizontally and $1/T_s$ vertically, every 2π.
Alias of f above $f_s/2$: fold about multiples of f_s	Example: 50 Hz at f_s=80 Hz lands at 30 Hz.	Noise outside the band can still alias onto the signal.
Modulated signal: compute the bandwidth of the signal you actually sample	Modulation by $\cos(\omega_c\, t)$ raises the bandwidth to $W+\omega_c$.	

Stuck? Try: Draw $X(\omega)$ and its copies every $2\pi/T_s$. Overlap means aliasing.

W9 Anti-aliasing and reconstruction

Use when: Reconstruction filter, interpolation, ZOH vs FOH vs sinc.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Always low-pass filter before sampling or downsampling	Cut off at the band you care about, not the raw bandwidth.	Skipping it is the most common project mistake.
Reconstruction: $y = h * w$, $w(t) = \Sigma\, x_s[n]\delta(t-nT_s)$, $Y(\omega) = H(\omega)X_s(T_s\omega)$		
Interpolation condition: $h(nT_s) = \delta[n]$	Both $h(0)=1$ and zeros at other sample times.	
Ideal: $H = T_s$ for $\lvert\omega\rvert<\pi/T_s$, else 0. $h(t) = \sin(\pi t/T_s)/(\pi t/T_s)$	$W \cdot T_s < \pi$.	Non-causal. Needs all past and future samples.
Shannon–Nyquist: $\omega_s > 2W \Rightarrow$ exact reconstruction by the sinc filter	Bandlimited signal.	ECG with 40 Hz bandwidth needs $f_s > 80$ Hz.
ZOH: $h = u(t) - u(t-T_s)$, delay $\approx T_s/2$. Causal FOH: delay T_s	Causal.	Better interpolation costs either causality or delay.
Signal not exactly bandlimited loses energy in the filter	Lost energy = $(1/2\pi)\int\lvert X\rvert^2$ over the removed band.	Use Parseval, not inspection.
Reconstruction recipe: put a scaled copy of h at each sample and sum		

W10 Laplace transform and RoC

Use when: Transform a signal, find the region of convergence.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\hat{X}(s) = \int x(t)e^{-st}dt$, $s = \sigma+j\omega$	The answer is incomplete without the RoC.	
RoC depends only on $\sigma = \mathrm{Re}(s)$	It is a vertical strip, half-plane, whole plane or empty. Never depends on ω.	
Right-sided ⇒ half-plane to the RIGHT of the rightmost pole	$e^{-\alpha t}u(t)$ and $-e^{-\alpha t}u(-t)$ share $1/(s+\alpha)$. Only the RoC tells them apart.	
Left-sided ⇒ half-plane to the LEFT of the leftmost pole		
Two-sided ⇒ vertical strip (or empty)		
Finite duration and absolutely integrable ⇒ whole plane		
RoC never contains a pole	Rational $\hat{X}$.	Use it to sanity-check any answer.
x has a CTFT iff the RoC contains the $j\omega$ axis. Then $X(\omega) = \hat{X}(j\omega)$	Absolutely integrable.	Right-sided with rightmost pole at $-a$: need $a>0$.
Rational: poles = roots of denominator, zeros = roots of numerator	Degree numerator > denominator ⇒ pole at infinity.	Check the discriminant when you find complex roots.
Sum or convolution: RoC ⊇ intersection	Pole–zero cancellation can enlarge it.	

Stuck? Try: Ask: is the signal right-, left- or two-sided? That fixes the RoC shape. Then the poles fix the edge.

W10 Laplace pairs and properties (given tables)

Use when: Combine rows. State the RoC every time.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\delta(t-\tau) \Leftrightarrow e^{-s\tau}$, RoC $= \mathbb{C}$		
$u(t) \Leftrightarrow 1/s$, $\mathrm{Re}(s)>0$		
$e^{-\alpha t}u(t) \Leftrightarrow 1/(s+\alpha)$, $\mathrm{Re}(s) > -\mathrm{Re}(\alpha)$		
$-e^{-\alpha t}u(-t) \Leftrightarrow 1/(s+\alpha)$, $\mathrm{Re}(s) < -\mathrm{Re}(\alpha)$		
$t^{n-1}/(n-1)!\, e^{-\alpha t}u(t) \Leftrightarrow 1/(s+\alpha)^n$, $\mathrm{Re}(s) > -\mathrm{Re}(\alpha)$		
$\cos(\omega_0 t)u(t) \Leftrightarrow s/(s^2+\omega_0^2)$. $\sin(\omega_0 t)u(t) \Leftrightarrow \omega_0/(s^2+\omega_0^2)$	$\mathrm{Re}(s)>0$.	
Linearity, convolution: RoC $\supseteq$ RoC$_1 \cap$ RoC$_2$		
Delay: $x(t-\tau) \Rightarrow e^{-s\tau}\hat{X}(s)$, same RoC		
Multiply by $e^{s_0 t}$: $\Rightarrow \hat{X}(s-s_0)$, RoC shifted by s_0		
Time scaling: $x(at) \Rightarrow (1/\lvert a\rvert)\hat{X}(s/a)$		
Multiply by t: $tx(t) \Rightarrow -d\hat{X}/ds$	The factor t becomes a derivative in s. Do not write a (s+2) factor.	
Integration: $\int x \Rightarrow \hat{X}/s$. Differentiation: $dx/dt \Rightarrow s\hat{X}(s)$	RoC $\supseteq$ RoC(x) (and $\mathrm{Re}(s)>0$ for integration).	Used for differential equations in Topic 11.

W11 Transfer function

Use when: LTI system in the Laplace domain. Differential equation to H(s).

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$x=e^{st} \Rightarrow y=\hat{H}(s)e^{st}$. $\hat{H}(s) =$ Laplace transform of h	$\hat{H}(s)$ with its RoC defines the system.	
$\hat{Y}(s) = \hat{H}(s)\hat{X}(s)$, $\hat{H} = \hat{Y}/\hat{X}$		
$H(\omega) = \hat{H}(j\omega)$	Stable (or jω axis in the RoC).	
Delay: $e^{-\tau s}$. Differentiator: s. Integrator: $1/s$ $(\mathrm{Re}(s)>0)$	Delay is not rational, so no partial fractions.	
ODE $\rightarrow \hat{H}$: replace d/dt by s, and x, y by $\hat{X}$, $\hat{Y}$	Initial rest.	RC circuit: $\hat{H} = (1/RC)/(s + 1/RC)$, one pole at $-1/RC$.
Series: $\hat{H}_1\hat{H}_2$. Parallel: $\hat{H}_1+\hat{H}_2$. Feedback: $\hat{H}_1/(1 - \hat{H}_1\hat{H}_2)$	Negative feedback means $\hat{H}_2 = -1$ (sign flips the denominator to $1+\hat{H}_1$).	

W11 RoC, causality and stability

Use when: "Is it causal / BIBO stable / both?" from poles.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Causal $\Leftrightarrow$ RoC is a right half-plane (for rational $\hat{H}$ also needs deg num $\leq$ deg den)		
BIBO stable $\Leftrightarrow$ RoC contains the $j\omega$ axis	Causal and stable are independent checks.	
Causal and BIBO stable (rational) $\Leftrightarrow$ all poles in the open left half-plane	A pole on the $j\omega$ axis (integrator) is not stable.	
Single pole at $-1/RC$ with $R,C>0$ is stable		

Stuck? Try: Locate poles. Decide the RoC from causality. Then ask whether $j\omega$ is inside it.

W11 Inverse Laplace by partial fractions

Use when: Find y(t) for a given input, or h(t) from $\hat{H}$.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$Y = HX$, then expand $\hat{Y}$ in partial fractions	Rational $\hat{Y}$.	RoC decides the time signal. Right of the pole $\Rightarrow e^{pt}u(t)$. Left $\Rightarrow -e^{pt}u(-t)$.
Strictly proper, simple poles: $\hat{Y} = \Sigma\, A_i/(s-p_i)$, $A_i = (s-p_i)\hat{Y}(s)$ at $s=p_i$	deg num < deg den.	
Not strictly proper: divide polynomials first	deg num $\geq$ deg den.	The quotient gives $\delta(t)$ and derivative terms.
Repeated pole: terms $1/(s-p)$, $1/(s-p)^2$, ...	$1/(s+\alpha)^n \Leftrightarrow t^{n-1}/(n-1)!\, e^{-\alpha t}u(t)$.	
Final value (step input, stable): $y(\infty) = \hat{H}(0)$		

Stuck? Try: Write $\hat{Y}$, factor the denominator, expand, then look up each term with the RoC you decided.

W12 Z-transform and RoC

Use when: DT version of Laplace. Transform of a sequence.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\hat{X}(z) = \Sigma\, x[n]\, z^{-n}$	Always state the RoC.	
RoC is a ring $\lvert z\rvert$ in (r_1, r_2)	Depends only on $\lvert z\rvert$.	Annulus must have inner < outer, else empty.
Right-sided ⇒ outside the outermost pole ($\lvert z\rvert > r$)	$z/(z-\alpha)$ is $\alpha^n u[n]$ ($\lvert z\rvert>\lvert\alpha\rvert$) and $-\alpha^n u[-n-1]$ ($\lvert z\rvert<\lvert\alpha\rvert$). Only the RoC differs.	
Left-sided ⇒ inside the innermost pole. Two-sided ⇒ annulus		
Finite duration ⇒ whole plane except $z=0$ (and ∞)	Negative powers of z blow up only at 0.	
DTFT $= \hat{X}(e^{j\omega})$ iff the RoC contains the unit circle	x absolutely summable.	A signal can have a Z-transform and no DTFT.
Left half-plane of s $\leftrightarrow$ inside the unit circle. $j\omega$ axis $\leftrightarrow$ unit circle	Stability: poles inside the unit circle (causal).	
RoC never contains a pole	Rational $\hat{X}$.	

W12 Z pairs and properties (given tables)

Use when: Combine rows. Write the RoC.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\delta[n] \Leftrightarrow 1$, RoC $\mathbb{C}$. $\delta[n-m] \Leftrightarrow z^{-m}$		
$u[n] \Leftrightarrow 1/(1-z^{-1})$, $\lvert z\rvert>1$		
$\alpha^n u[n] \Leftrightarrow 1/(1-\alpha z^{-1})$, $\lvert z\rvert>\lvert\alpha\rvert$		
$\alpha^n u[-n] \Leftrightarrow 1/(1-\alpha^{-1}z)$, $\lvert z\rvert<\lvert\alpha\rvert$	Note $u[-n]$, not $u[-n-1]$.	
$n\alpha^n u[n] \Leftrightarrow \alpha z^{-1}/(1-\alpha z^{-1})^2$, $\lvert z\rvert>\lvert\alpha\rvert$		
$\alpha^n \cos(\omega_0 n)u[n] \Leftrightarrow (1-\alpha z^{-1}\cos\omega_0)/(1-2\alpha z^{-1}\cos\omega_0+\alpha^2 z^{-2})$	$\lvert z\rvert>\lvert\alpha\rvert$	
$\alpha^n \sin(\omega_0 n)u[n] \Leftrightarrow \alpha z^{-1}\sin\omega_0/(1-2\alpha z^{-1}\cos\omega_0+\alpha^2 z^{-2})$	$\lvert z\rvert>\lvert\alpha\rvert$	
Delay: $x[n-m] \Rightarrow z^{-m}\hat{X}(z)$, RoC minus $\{0\}$		
Modulation: $\alpha^n x[n] \Rightarrow \hat{X}(z/\alpha)$, RoC $= \alpha\cdot$RoC		
Convolution: $x_1 * x_2 \Rightarrow \hat{X}_1\hat{X}_2$, RoC $\supseteq$ intersection		
Time reversal: $x[-n] \Rightarrow \hat{X}(z^{-1})$		
Conjugation: $x^* \Rightarrow [\hat{X}(z^*)]^*$		

W12 DT transfer function and stability

Use when: Difference equation to H(z), poles, step response.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\hat{H}(z)$: replace $x[n-m]$ by $z^{-m}\hat{X}$, divide $\hat{Y}$ by $\hat{X}$	Initial rest = causal = RoC outside the outermost pole.	Read $\hat{H}$ straight off the equation.
$H(\omega) = \hat{H}(e^{j\omega})$	RoC contains the unit circle.	
Causal $\Leftrightarrow$ RoC outside a circle. BIBO stable $\Leftrightarrow$ RoC contains the unit circle	Independent checks.	Accumulator $y[n]=y[n-1]+x[n]$ has $\hat{H} = z/(z-1)$: causal, pole on the unit circle, not stable.
FIR (all zeros, poles only at 0) is always stable		
Series: $\hat{H}_1\hat{H}_2$. Parallel: $\hat{H}_1+\hat{H}_2$. Feedback: $\hat{H}_1/(1-\hat{H}_1\hat{H}_2)$		
Inverse: expand $\hat{Y}/z$ in partial fractions, multiply back by z	Use RoC to choose right or left sided terms.	Repeated pole $\Rightarrow n\alpha^n$-type term.
Step response final value $= \hat{H}(1)$	Stable.	

Stuck? Try: Write $\hat{H}$ in z, factor, check pole magnitudes against 1, then partial fractions on $\hat{Y}/z$.

Built from your ECE2111 weekly notes (W1 to W12; no W7 supplied) and the 2022 and 2024 exams, sample paper, mock exam and mid-sem test. Conventions follow the course: ω in rad/s, CTFT inverse has 1/2π, u(0) left undefined. A few textbook standard results (square-pulse series coefficients, ideal low-pass impulse response) are included; check them against your lecture tables.